MIL-R-9405A(USAF)
29 April 1957

Superseding
MIL-R-9405(USAF)
4 January 1954

MILITARY SPECIFICATION

RELAY ASSEMBLY RE-94A/AIC-10

1. SCOPE

1.1 This specification covers one type of relay assembly designated Relay Assembly RE-94A/AIC-10. This relay assembly includes AF Amplifier AM-476( )/AIC-10 and is one item of Intercommunication Set AN/AIC-10A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-R-11	Resistors, Fixed, Composition (Insulated)
MIL-A-5144	Automatic Gain Control C-158A/AIC
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-E-5557	Enamel; Heat-Resisting, Glyceryl-Phthalate, Black
MIL-E-5558	Enamel; Wrinkle-Finish, For Aircraft Use
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-T-9107	Test Reports, Preparation Of
MIL-A-9286	AF Amplifier, AM-476A/AIC-10
MIL-W-16878	Wire, Electrical (Insulated, High Temperature)
MIL-E-17555	Preparation For Delivery of Electronic Equipment: Miscellaneous Electrical Equipment (Except Rotating Equipment) and Associated Repair Parts

STANDARDS

Army-Navy Aeronautical

AN3102	Connectors - Electrical, Receptacles, Box Mounting

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DRAWINGS

U. S. Air Force

52C12817	Wiring Diagram, Relay Assembly RE-94A/AIC-10, Schematic
53D13999	Wiring Diagram, Intercommunication Set AN/AIC-10A, "Hot Mic System", Test Circuit, Schematic
53D13658	Relay Assembly RE-94A/AIC-10, Outline Dimensions Of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The requirements of MIL-E-5400 for class 2 equipment apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.1.1 Substitution of Material.- If a manufacturer, either prior to or during production, desires to substitute another material or process for one previously approved or specified, he shall submit to the procuring activity a statement to that effect. The statement shall describe the proposed substitution and the reason therefor, together with evidence to show that the substitution is satisfactory. At the discretion of the procuring activity, test samples or additional data may be required to prove the adequacy of the proposed substitution.

3.2 Standard Model.- A model of Relay Assembly RE-94A/AIC-10 shall serve as a standard of design, materials, and performance only in those instances not specifically covered by this specification. It is required that all major component parts of the relay assembly be completely interchangeable electrically and mechanically with the corresponding parts of the model equipment. Departure from the model equipment design can be made only when specifically permitted by the procuring activity.

3.3 Design and Construction

3.3.1 Drainage.- Each relay assembly shall incorporate adequate drainage provisions for any moisture condensing within the unit. When the relay assembly is mounted in any usable position, including horizontal, oblique, vertical, overhead, and sidewise mounting, all

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closed corners, channels, etc, which would collect and hold water shall be eliminated or be provided with adequate means for drainage. A properly located round hole of one-eighth inch diameter is considered satisfactory drainage for a corner in a sheet metal box.

3.3.2 Circuitry.- The relay assembly shall be wired in accordance with Drawing No. 52C12817.

3.3.3 Wire.- The hookup wire shall be AWG No. 22 (642.5 circular mils) in accordance with MIL-W-16878, type B. The outer covering shall be a synthetic textile or glass braid. Color coding shall be like that of the model relay assembly.

3.3.4 Cabling.- In general, the cabling shall be patterned after that of the model equipment. Each cable harness shall be fabricated in such a manner that interaction between any combination of wires or components shall not cause hum, stray pickup, or oscillation.

3.3.5 Relays.- Relays used shall be either RCA Type 459143 or Struthers-Dunn Type FC-6-110, or equal.

3.3.6 Barometric Switches.- The barometric switches shall be no larger or heavier than those in the model relay assembly. The performance of the barometric switches shall be in accordance with MIL-A-5144, Automatic Gain Control C-158A/AIC. The nominal operating altitude (that altitude at which the switch closes) of one barometric switch shall be 20,000 feet (13.74 inches Hg) and that of the other barometric switch shall be 30,000 feet (8.88 inches Hg).

3.3.7 Filter Capacitor.- The filter capacitor shall be a 40 or 60 microfarad, 50 volt, tantalum capacitor.

3.3.8 Resistors, Fixed.- The fixed resistors shall be in accordance with table I.

TABLE I

Resistor	Ohms	Watts	Type	Specification
R-201	220	1/2	RC20BF221K	MIL-R-11
R-202	390	1/2	RC20BF391K	MIL-R-11

3.3.9 Electrical Receptacle.- The external electrical receptacle shall be Type AN3102A-20-27P.

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3.3.10 AF Amplifier AM-476( )/AIC-10A.- AF Amplifier AM-476( )/AIC-10A shall be in accordance with MIL-A-9286.

3.3.11 Circuit Label.- The circuit label shall be similar to the schematic diagram shown on Drawing No. 52C12817 and shall include the work identification of the terminals. The complete nomenclature of the relay assembly shall be used to identify the schematic. A sketch of the proposed circuit label shall be submitted to the procuring activity for approval prior to fabrication of the production circuit labels.

3.4 Electrical Performance

3.4.1 Operation with Intercommunication Set AN/AIC-10A.- The relay assembly shall provide satisfactory performance and shall not adversely affect the performance of any other component of Intercommunication Set AN/AIC-10A, when interconnected with and operated as a part of that equipment.

3.4.2 Radio Interference.- The relay assembly shall meet the radio noise interference requirements of MIL-I-6181.

3.4.3 Noise in Output Due to Ripple on DC Line.- With a 1.5 volt (rms) ripple voltage superimposed on the 27.5 volt dc line, the noise voltage across a 150-ohm resistor connected across the relay assembly output (with the barometric switches open) shall not exceed the values indicated in table II.

TABLE II

Ripple Frequency (cps)	Maximum Noise (volts)
400	0.0056
1200	0.011
6000	0.04

3.4.4 Output-to-Input Feedback.- A capacitance of 0.004 mfd connected from the output of the relay assembly (with the barometric switches shorted) to either of the microphone inputs (with the associated relay energized) shall not cause the amplifier to oscillate.

3.4.5 Crosstalk.- Crosstalk in the output circuit resulting from an input signal on either microphone input circuit with the associated relay unenergized shall be at least 50 db down from the output level resulting when the associated relay is energized.

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3.4.6 Output with 5-Ohm Source.- With a 1000 cps input of 0.075 volt (rms) into a standard 5-ohm pad (per MIL-A-9286) coupled to one ~~microphone input circuit (with its relay energized)~~ the output of the relay assembly, measured across a 150-ohm load, with the barometric switches shorted, shall be at least 0.291 volts (rms).

3.4.7 Output with 2.5-Ohm Source.- With a 1000 cps input of 0.1 volt (rms) into a standard 2.5-ohm pad (per MIL-A-9286) coupled to either of the two energized microphone input circuits, the output of the relay assembly measured across a 150-ohm load, with the barometric switches shorted, shall be at least 0.19 volts (rms).

3.4.7.1 When the short is removed from the 30,000 foot barometric switch, the output shall be reduced by 5 db, ±1.5 db. When the shorts are removed from both the 30,000 foot and 20,000 foot switches, the output shall be reduced by 10 db, ±1.5 db.

3.5 Dimensions.- The overall dimensions of the relay assembly shall not exceed those shown on Drawing No. 53D13658.

3.6 Weight.- The weight of the relay assembly, including AF Amplifier AM-476( )/AIC-10A, shall not exceed 2.1 pounds.

3.7 Current Drain.- With both relays and the amplifier energized, the 27.5 volts dc current drain shall not exceed 400 milliamperes.

3.8 Finish.- The exterior surfaces of the relay assembly shall be furnished with black wrinkle enamel in accordance with MIL-E-5558. The interior of the housing shall be finished with dull black enamel in accordance with MIL-E-5557 except where films or coatings may interfere with essential electrical contacts or grounding points.

3.9 Government-loaned Property.- If available, and when provided for in the contract or purchase order, the following items will be furnished by the Government on loan to the contractor upon his request:

Item No.	Stock Number	Item Description	Quantity
1	1600 013635000	Relay Assembly RE-94A/AIC-10	1
2	1600 211945170	Control Panel C-824A/AIC-10	2
3	1790 207385-29	Headset-Microphone H-78( )/AIC	2

3.9.1 The Government-loaned equipment shall not be damaged or disassembled except where the assembly is attached with machine screws, nor shall the equipments so furnished be used as a criteria of performance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.- The inspection and testing of the relay assembly shall be classified as follows:

a. Preproduction tests.....See 4.3

b. Acceptance tests.....See 4.4

4.2 Test Conditions.- Unless otherwise specified, the relay assembly shall be tested under the following conditions:

Ambient room temperature, plus 20°C to plus 35°C
Relative humidity, prevailing room ambient (20 to 90 percent relative)
Atmospheric pressure, prevailing ground level ambient
Primary voltage supply, 27.5 volts dc

4.3 Preproduction Tests.- Preproduction tests shall be performed by the contractor in accordance with the following:

4.3.1 Preproduction Test Samples.- Preproduction test samples shall consist of the following:

4.3.1.1 Six each Relay Assembly RE-94A/AIC-10, when specified in the contract, shall be subjected by the contractor to the preproduction tests specified herein.

4.3.1.2 Two each Relay Assembly RE-94A/AIC-10, when specified in the contract, shall be submitted to the procuring activity at the time the preproduction sample test report is submitted for approval. The samples will be used by the procuring activity for:

a. A review of the mechanical construction of the relay assembly.

b. Performance of any tests which may be deemed necessary after a review of the contractor's test report.

c. Government reference purposes.

4.3.2 Preproduction Test Report.- After completion of the preproduction tests by the contractor, three copies of a complete report shall be furnished the procuring activity. The preproduction test report shall be in accordance with MIL-T-9107. In addition, the test report shall include the name and grade of all materials used in the relay assembly.

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4.3.3 Preproduction Test Program.- Prior to any formal testing of the preproduction relay assemblies, the contractor shall submit a draft of his proposed test program to the procuring activity for approval. This should be done as early as possible to prevent any delays in the preproduction testing of relay assemblies. The draft of the proposed test program shall include:

a. A list of all tests to be performed, and complete procedures for each test including block or schematic diagrams.

b. A list of test equipment to be used, identified by manufacturer, and type number.

c. Copies of the data record forms to be used in recording the test data.

d. Schematics of the proposed test sets, and test set operating instructions.

4.3.4 Preproduction Test Routine.- Preproduction samples shall be subjected to the test listed in table III in the quantities shown. AF Amplifier AM-476( )/AIC-10 shall be subjected to the preproduction tests specified in MIL-A-9286 prior to its assembly into, and testing of, the complete RE-94A/AIC-10 assembly.

4.4 Acceptance Tests.- Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests

4.4.1 Individual Tests.- Each relay assembly shall be subjected to the individual tests listed in table III in the quantities shown. AF Amplifier AM-476( )/AIC-10 shall be subjected to the individual tests specified in MIL-A-9286 prior to its assembly into, and testing of, the complete RE-94A/AIC-10 assembly.

4.4.2 Sampling Tests.- Sampling tests shall be made on at least one percent of the relay assemblies on order. The relay assemblies shall be selected at random from quantities of relay assemblies which have passed individual tests. The samples shall be subjected to the tests listed in table III in the quantities shown. AF Amplifier AM-476( )/AIC-10 shall be subjected to the sampling tests specified in MIL-A-9286.

4.4.2.1 Acceptance Criteria.- When one or more of the selected relay assemblies fails to meet the specification, acceptance of all items will

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be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

TABLE III

Test	Paragraph or Spec.	Preproduction	Individual	Sampling
(These tests need not be performed in order shown)				
A. Mechanical Inspection	4.5.1	all	all	all
B. Electrical Tests	4.5.2.2	---	---	---
Circuitry	4.5.2.2	all	all	all
Noise in output due to ripple on dc line	4.5.2.4	all	all (only at 3000 cps)	all
Output-to-input feedback	4.5.2.6	all	---	all
Output-with 5-ohm source	4.5.2.7	all	all	all
Output with 2.5-ohm source	4.5.2.8	all	all	all
Current Drain	4.5.2.5	all	---	all
Crosstalk	4.5.2.3	all	all	all
Operation with intercommunication Set AN/AIC-10A	4.5.2.9	all	---	all
Radio Noise Interference	MIL-I-6181	all	---	---
C. Environmental Tests	4.5.3	---	---	---
Fungus		all	---	---
Salt-spray		all	---	---
Humidity		all	---	10%
Vibration		all	---	10%

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TABLE III - Cont'd

Test	Paragraph or Spec.	Preproduction	Individual	Sampling
Shock		all	---	---
High & low temperature		all	---	10%
Explosion proof		all	---	---
Altitude		all	20%	10%

4.5 Test Procedures

4.5.1 Mechanical Inspection.- The relay assembly shall be examined to determine compliance with the requirements of this specification for materials, workmanship, dimensions, and markings using whatever tools, gauges, scales, and the like that are necessary. Fastening devices shall be checked for proper mechanical operation.

4.5.2 Electrical Tests

4.5.2.1 Input-Output Terminations.- Where required, the input pads shall be as specified in MIL-A-9286 for testing AF Amplifier AM-476( )/AIC-10. The audio output line shall be terminated with a 150-ohm resistor.

4.5.2.2 Circuitry.- The relay assembly shall be tested for continuity of wiring, correctness of wiring and component parts, barometric switch and relay operation. The test circuit used to accomplish the above shall be such as to detect improper grounds and short circuits.

4.5.2.3 Crosstalk.- Crosstalk measurements shall be performed for the following operating conditions using a 0.3 millivolt (rms) 1000 cps input signal:

a. Inject the input signal into MIC INPUT NO. 1 (relay energized) and measure the output level across the 150-ohm load.

b. Inject the input signal into MIC INPUT NO. 1 (relay unenergized) and measure the crosstalk level across the 150-ohm output load.

c. Repeat step a. and b. for MIC INPUT NO. 2.

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4.5.2.4 Noise in Output Due to Ripple on DC Line.- Using a 1.5 volt (rms) input signal, noise measurements shall be performed for each condition of operation, specified in 3.4.3.

4.5.2.5 Current Drain.- Measurements shall be made to determine that that 27.5 volts dc current drain, under any mode of operation, conforms to 3.7.

4.5.2.6 Output-to-Input Feedback.- This test shall be performed with one microphone input circuit loaded with a 5-ohm resistor and with its associated microphone relay circuit energized. The output circuit shall be loaded with a 150-ohm resistor during this test.

4.5.2.7 Output with 5-Ohm Source.- The relay assembly shall be tested for compliance with the requirements of 3.4.6 in accordance with the conditions specified therein.

4.5.2.8 Output with 2.5-Ohm Source.- The relay assembly shall be tested to determine its output under each mode of operation specified in 3.4.7 and 3.4.7.1.

4.5.2.9 Operation with Intercommunication Set AN/AIC-10A.- The relay assembly under test shall be connected into a two-place hot mic AN/AIC-10A interphone system in accordance with the wiring diagram shown on Drawing No. 50D13999. The relay assembly shall then be subjected to the tests specified on that drawing.

4.5.3 Environmental Tests.- The environmental tests listed in table IV shall be performed in accordance with the specifications indicated, and with the additional instructions given below:

TABLE IV

Test	Specification	Procedure
*Fungus	MIL-E-5272	I
*Salt spray	MIL-E-5272	I
*Humidity	MIL-E-5272	II
*Vibration	MIL-E-5272	XI
*High temperature	MIL-E-5272	I
Low temperature	MIL-E-5272	II
*Shock	MIL-E-5272	II
Explosion proof	MIL-E-5272	I
Altitude	4.5.3.4	---

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4.5.3.1 The circuitry and the talk-out tests specified in 4.5.2.9 shall be performed before and after each environmental test. The tests in table IV marked with asterisks shall be followed by altitude retests.

4.5.3.2 Vibration.- The relay assembly shall be attached to the vibration table using the mounting holes provided in the box. The amplitude of vibration shall be measured on the same surface to which the relay assembly is secured. The relay assembly shall be connected as an operating unit during the vibration tests. Relay bounce and chattering shall be checked during the shock tests by means of an oscilloscope connected to the output terminals of the relay assembly.

4.5.3.4 Altitude.- The altitude test shall be conducted as follows: The relay assembly shall be placed in an altitude chamber and the chamber shall be evacuated to a pressure equivalent to 55,000 feet altitude and then the pressure in the chamber shall be increased to normal barometric pressure. Repeat this portion of the test five times. By means of an ohmmeter or other indicating device, establish the altitude (pressure) at which each barometric switch closes and opens. The rate of ascent or descent shall be maintained at less than 2000 feet per minute. Perform the switch make and break tests a minimum of three times.

5. PREPARATION FOR DELIVERY

5.1 General.- Preservation, packaging, packing, and marking shall be in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use.- Relay Assembly RE-94A/AIC-10 is a switching device intended for use in military aircraft as a part of AN/AIC-10A hot-mic interphone systems. The RE-94A/AIC-10 operates only in conjunction with two each Control Panel C-824A/AIC-10. Hot mic systems are systems which enable users to talk and listen on the interphone channel without the necessity for operating a push-to-talk switch.

6.2 Ordering Data.- Invitation for bids, contracts, and purchase orders should state the conditions for the following:

6.2.1 Preproduction Tests.- It is expected that the contract or purchase order will specify that a minimum of six relay assemblies will be required as preproduction samples to be subjected to preproduction tests by the contractor to determine compliance with the requirements of the specification and that two each Relay Assembly RE-94A/AIC-10 will be required as preproduction samples to be submitted to the procuring activity.

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6.3 Model Equipment.- The model shall be maintained in its original condition, however, the contractor may loosen parts or subassemblies held in place by means of screws in order to examine the wiring. The soldered connections should not be disturbed nor should the model be modified.

6.4 Manufacturer's Drawings.- Manufacturer's drawings of Relay Assembly RE-94A/AIC-10 produced on the first production contract for this item are available from the procuring activity and may be used for informational purposes.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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